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Search terms

Pubmed search terms:

((connectivity[Title/Abstract] OR network[Title/Abstract] OR networks[Title/Abstract] OR subnetwork[Title/Abstract] OR subnetworks[Title/Abstract] OR graph[Title/Abstract] OR graphs[Title/Abstract] OR graph OR disconnection[Title/Abstract] OR dysconnection[Title/Abstract] OR connectome[Mesh] OR connectome[Title/Abstract] OR network[Title/Abstract] OR networks[Title/Abstract] OR subnetwork[Title/Abstract] OR subnetworks[Title/Abstract] OR dysconnection[Title/Abstract] OR connectomics[Title/Abstract] OR connectomic[Title/Abstract] OR neural communication[Title/Abstract] OR FC)) **AND** (spanning tree*[Title/Abstract] OR spanning graph*[Title/Abstract] OR MST[Title/Abstract] OR MWST)

Embase search terms:

(connectivity:ab,ti OR graph:ab,ti OR graphs:ab,ti OR disconnection:ab,ti OR connectome:ab,ti OR connectomes:ab,ti OR network:ab,ti OR networks:ab,ti OR subnetwork:ab,ti OR subnetworks:ab,ti OR connectomic:ab,ti OR connectomics:ab,ti OR 'neural communication':ab,ti OR fc:ab,ti) **AND** ('spanning tree':ab,ti OR 'spanning graph':ab,ti OR mst:ab,ti OR mwst:ab,ti)

Supplementary table 1: Overview of included studies

Category	*First Author*	*Clinical group*	*N*	*Modality*	*Nodes*
Neurodevelopmental disorders	*Janssen 2017*	Children with ADHD	42	EEG	78
		Typically developing children	43		
	Wang 2020	Children with ADHD	102	fMRI	264
		Typically developing children	143		
	González 2016	Dyslexic children	29	EEG	64
		Typically reading children	15		
	González 2018	Dyslexic adults	28	EEG	64
		Typical readers	36		
	Xue 2020	Dyslexic children	27	EEG	30
		Typically reading children	40		
	Zare 2016	Infants with a family history of language learning disorder	12	EEG	62
		Typically developing infants	12		
	Zeng 2017	High-functioning autism children	21	EEG	128
		Typically developing children	21		
Adult psychiatric disorders	*Krukow 2019*	Schizophrenia patients	35	EEG	19
		Healthy control	35		
	Jonak 2019	First episode psychosis	30	EEG	21
		Long duration illnes psychosis	30		
	Van Dellen 2020	Bipolar-I disorder	136	fMRI	264
		Schizophrenia spectrum patients	97		
		Sub-clinical psychosis	35		
		Healthy control	219		
	Anjomshoa 2016	Schizophrenia patients	27	DTI	116
		Healthy control	19		
	Van Dellen 2016	Sub-clinical psychosis	35	DTI	82
		Psychotic disorder	35		
		Healthy control	36		

	Li *2017*	Major depressive disorder Healthy control	23 15	EEG	72
	Wang 2019	Internet addiction Healthy control	30 30	EEG	64
Neurodegenerative diseases	*Van Dellen* *2015*	Lewie body dementia Alzheimers Disease Control	66 66 66	EEG	19
	Olde Dubbelink *2014*	Parkinsons disease *Parkinsons disease de novo (subgroup)* Controls	43 *12* 14	MEG	78
	Yu *2016*	Alzheimers Frontotemporal dementia Subjective cognitive decline	69 48 64	EEG	21
	Das *2020*	Mild cognitive impaired Alzheimers Healthy controls	13 20	EEG	23
	Utianski *2016*	Parkinsons disease cognitively normal Parkinsons disease MCI Parkinsons disease dementia Healthy control	57 13 18 57	EEG	21
	Saba *2019*	Frontotemporal Dementia Healthy control	41 39	fMRI	116
	Peraza *2018*	Alzheimers disease Dementia with Lewy bodies Parkinson's disease dementia Healthy controls	16 25 21 17	EEG	128
	Pozar *2020*	Mild cognitive impairment Cognitively normal	13 27	EEG	64
	Fraschini *2016*	Amyotrophic lateral sclerosis Healthy controls	21 16	EEG	58
	Sorrentino *2018*	Advanced amyotrophic lateral sclerosis Early amyotrophic lateral sclerosis	24 26	MEG	90

		Healthy controls	25		
	Sorrentino 2017	Subjects from AGHLS	156	MEG	80
	Jacini 2018	Amnestic mild cognitive impairment Healthy control	16 16	MEG	90
	Lopez 2017	Mild cognitive impairment Healthy control	29 29	MEG	78
	Wang 2018	Alzheimer's disease Mild cognitive impairment Healthy control	27 62 29	fMRI	90
	Jonak 2021	LHON Healthy controls	15 17	DTI	82
Multiple sclerosis	*Tewarie 2014 (1)*	Multiple sclerosis Healthy control	21 17	MEG	78
	Tewarie 2015	Multiple sclerosis Healthy control	86 21	MEG/fMRI	78
	Tewarie 2014 (2)	Multiple sclerosis Healthy control	102 42	MEG	78
	Nauta 2020	Multiple sclerosis/ follow-up	146 /100	MEG	78
Epilepsy	*Fraschini 2014*	Pharmaco resistant epilepsy Nonresponder to VNS Responder to VNS	 5 5	EEG	19
	Van Diessen 2016	Drug naive chld generalized epilepsy Drug naive chld focal epilepsy Healthy control	27 62 179	EEG	17
	Van Dellen 2014	Meningioma with seizures *Seizure free post surgery* *Seizures post surgery*	 13 7	MEG	78
	Van Diessen 2014	chld. suspected new onset focal epilepsy Control	21 17	EEG	21

		sleep deprevation (design)			
	Kinney-Lang 2019	Children with early onset epilepsy Cognitively normal Mild cognitive impairment Severe cognitive impairment	 31 7 13	EEG	20
	DeSalvo 2020	Temporal lobe epilepsy (TLE) TLE pre-op, seizures post-op TLE pre-op, seizure free post-op	40	fMRI	83
Other	*Van Montfort 2018*	Delirium Post Delirium Non delirious control	9 7 13	fMRI	90
	Numan 2017	Hypo-active delirium Recovering from anesthesia Non delirious control	18 20 20	EEG	17
	Nieboer 2020	Migraine *Recently diagnosed (subgroup)* *Subgroup long-term illnes (subgroup)* Healthy control	24 *12* *12* 24	MEG	78
	Van Nieuwenhuizen 2018	Meningioma patients Healthy control	20 20	MEG	78

Included articles are arranged by disease category. Nodes: the number of nodes in the network.

ADHD: attention deficit hyperactivity disorder, LLD: language learning disorder, MCI: mild cognitive impairment, AGHLS: Amsterdam Growth and Health Longitudinal Study, LHON: Leber's hereditary optic neuropathy, VNS: vagal nerve stimulation

Supplementary table 2: MST metrics from studies on Neurodevelopmental disorders

Modality	***First Author***	**Clinical group**	**Diameter**	**LF**	**Kappa**	**Th**
EEG delta	*González 2016*	Dyslexic children	0.219 ± 0.014	0.576 ± 0.020	3.501 ± 0.270	0.417 ± 0.023
		TRC	0.216 ± 0.013	0.583 ± 0.012	3.551 ± 0.259	0.418 ± 0.015
	Zare 2016	Infants with a family history of LLD	N/A	N/A[a b]	N/A	N/A[a b]
		TDI	N/A	N/A [a b]	N/A	
EEG theta	*González 2016*	Dyslexic children	**0.224 ± 0.011**	**0.569 ± 0.015**	3.341 ± 0.165	0.412 ± 0.016
		TRC	**0.216 ± 0.015**	**0.584 ± 0.013**	3.415 ± 0.149	0.419 ± 0.015
	Janssen 2017	ADHD children	0.234 ± 0.018	0.491 ± 0.025	N/A	0.370 ± 0.020
		TDC	0.237 ± 0.018	0.482 ± 0.019	N/A	0.364 ± 0.019
	Xue 2020	Dyslexic children	0.360 ± 0.016	0.544 ± 0.016	2.920 ± 0.109	0.392 ± 0.010
		TRC	0.356 ± 0.015	0.545 ± 0.016	2.945 ± 0.099	0.390 ± 0.012
	Wang 2019	Internet addicition	**0.064** ± **0.009**	**0.957** ± **0.086**	**26.467** ± **8.301**	0.498 ± 0.025
		Control	**0.094 ± 0.009**	**0.919 ± 0.078**	**21.402 ± 8.743**	0.498 ± 0.026
EEG alpha	*González 2016*	Dyslexic children	0.206 ± 0.015	0.609 ± 0.027	3.892 ± 0.466	0.432 ± 0.024
		TRC	0.197 ± 0.018	0.623 ± 0.030	3.967 ± 0.398	0.441 ± 0.018
	Janssen 2017	ADHD children	**0.220 ± 0.018**	**0.508 ± 0.023**	N/A	**0.379 ± 0.018**
		TDC	**0.230 ± 0.023**	**0.494 ± 0.030**	N/A	**0.369 ± 0.023**
	Zeng 2017	High-functioning autism children	N/A	N/A[a]	N/A	N/S
		TDC	N/A	N/A [a]	N/A	N/S
	González 2018	Dyslexic adults	0.201 ± 0.012	0.623 ± 0.027	**4.061 ± 0.443**	0.437 ± 0.015
		Typical readers	0.206 ± 0.012	0.615 ± 0.024	**3.857 ± 0.289**	0.437 ± 0.015
EEG beta	*González 2016*	Dyslexic children	0.216 ± 0.012	0.580 ± 0.022	3.542 ± 0.275	0.417 ± 0.020
		TRC	0.216 ± 0.013	0.582 ± 0.018	3.516 ± 0.244	0.425 ± 0.015
	Janssen 2017	ADHD children	0.220 ± 0.023	**0.502 ± 0.030**	N/A	0.371 ± 0.023

		TDC	0.227 ± 0.019	**0.485 ± 0.027**	N/A	0.363 ± 0.019
	Xue 2020	Dyslexic children	0.359 ± 0.013	0.531 ± 0.015	2.863 ± 0.098	0.384 ± 0.012
		TRC	0.354 ± 0.012	0.538 ± 0.020	2.907 ± 0.125	0.383 ± 0.014
EEG gamma	*González 2016*	Dyslexic children	0.205 ± 0.019	0.621 ± 0.038	4.468 ± 1.381	0.431 ± 0.020
		TRC	0.196 ± 0.014	0.637 ± 0.034	4.587 ± 0.875	0.443 ± 0.020
fMRI	*Wang 2020*	ADHD children	N/A	**0.421 ± 0.023**	**1.670 ± 0.099**	**1.304 ± 0.103**
		TDC	N/A	**0.430 ± 0.024**	**1.706 ± 0.098**	**1.344 ± 0.115**

Data are presented as mean and when available SD. (SE) indicates the use of standard error instead of SD.

Bold text represents significant results ($p < 0.05$). For EEG only frequency bands for which data was available are displayed.

N/A: not available, measure not described in study; N/A [a] : significant difference for pathology compared to control group is described in study, but numeric data not available; N/S: not significant; [b] :comparison based on the leaf number, significant compared to controls,

EEG: electroencephalography, fMRI: functional magnetic resonance imaging, ADHD: Attention-deficit/hyperactivity disorder, TDC: typically developing children, TRC: typically reading children, TDI: typically developing infants, LLD: language-learning disorder.

Supplementary table 3: MST metrics from studies on adult psychiatric disorders

Modality	***First Author***	**Clinical group**	**Diameter**	**LF**	**Kappa**	**Th**
EEG delta	*Krukow 2019*	SCZ	N/A[a]	N/A[a]	N/A	N/A
		Control	N/A	N/A	N/A	N/A
	Jonak 2019	FES	0.416	0.564	N/A	0.358
		LDI	0.412	0.539	N/A	0.338
	Wang 2019	Internet addicition	0.131 ± 0.054	0.849 ± 0.100	12.478 ± 6.730	0.513 ± 0.058
		Control	0.119 ± 0.043	0.859 ± 0.098	12.043 ± 6.179	0.527 ± 0.071
EEG theta	*Li 2017*	Major depressive disorder	N/A	**0.598**	N/A	N/A
		Control	N/A	**0.544**	N/A	N/A
	Krukow 2019	SCZ	N/S	N/S	N/A	N/A
		Control	N/S	N/S	N/A	N/A
	Jonak 2019	FES	0.436	0.524	N/A	0.332
		LDI	0.396	0.531	N/A	0.324
	Wang 2019	Internet addicition	**0.064** ± **0.009**	**0.957** ± **0.086**	**26.467** ± **8.301**	0.498 ± 0.025
		Control	**0.094 ± 0.009**	**0.919 ± 0.078**	**21.402 ± 8.743**	0.498 ± 0.026
EEG lower alpha	*Krukow 2019*	SCZ	N/S	N/S	N/A	N/A
		Control	N/S	N/S	N/A	N/A
	Jonak 2019	FES	0.446	0,513	N/A	0.363
		LDI	0.420	0.563	N/A	0.353
EEG upper alpha	*Krukow 2019*	SCZ	N/S	N/S	N/A	N/A
		Control	N/S	N/S	N/A	N/A
	Jonak 2019	FES	0.432	0.516	N/A	0.327
		LDI	0.398	0.556	N/A	0.347
	Wang 2019	Internet addicition	**0.064** ± **0.009**	**0.957** ± **0.086**	**26.467** ± **8.301**	0.498 ± 0.025
		Control	**0.094 ± 0.009**	**0.919 ± 0.078**	**21.402 ± 8.743**	0.498 ± 0.026
EEG beta	*Krukow 2019*	SCZ	N/A[a]	N/S	N/A	N/A
		Control	N/A[a]	N/S	N/A	N/A

	Jonak 2019	FES	0.445	0.533	N/A	**0.330**
		LDI	0.399	0.606	N/A	**0.408**
	Wang 2019	Internet addicition	**0.066** ± **0.034**	0.961 ± 0.041	**25.014** ± **7.271**	0.509 ± 0.044
		Control	**0.090 ± 0.053**	0.928 ± 0.073	**19.480 ± 8.141**	0.520 ± 0.063
EEG gamma	*Krukow 2019*	SCZ	N/A[a]	N/A[a]	N/A	N/A
		Control	N/A[a]	N/A[a]	N/A	N/A
	Jonak 2019	FES	0.413	0.548	N/A	**0.373**
		LDI	0.399	0.553	N/A	**0.277**
DTI	*Anjomshoa 2016*	SCZ	**0.566***	**0.451***	**2.779**	N/A
		Control	**0.525***	**0.492***	**2.869**	N/A
	Van Dellen 2016	Sub-clinical psychosis	0.192 ± 0.029	0.492 ± 0.028	2.932 ± 0.180	0.361 ± 0.033
		Psychotic disorder	0.201 ± 0.026	0.483 ± 0.034	2.938 ± 0.214	0.359 ± 0.041
		Control	0.197 ± 0.029	0.491 ± 0.041	2.905 ± 0.163	0.3627 ± 0.037
fMRI	*Van Dellen 2020*	Bipolar-I disorder	0.111 ± 0.002 (SE)	**0.465 ± 0.002 (SE)**	**2.802 ± 0.002 (SE)**	N/A
		Schizophrenia spectrum patients	0.111 ± 0.002 (SE)	**0.471 ± 0.003 (SE)**	**2.835 ± 0.004 (SE)**	N/A
		Sub-clinical psychosis	0.108 ± 0.003 (SE)	**0.478 ± 0.004 (SE)**	**2.867 ± 0.003 (SE)**	N/A
		Control	0.108 ± 0.001 (SE)	**0.472 ± 0.002 (SE)**	**2.840 ± 0.008 (SE)**	N/A

Data are presented as mean and when available SD. (SE) indicates the use of standard error instead of SD. Bold text represents significant results ($p < 0.05$). N/A: not available: measure not described in study, N/A[a] significant difference for pathology compared to control group is described in study, but numeric data not available; N/S: not significant; *normalized values calculated from the raw values of the article.

EEG: electroencephalography, fMRI: functional magnetic resonance imaging, DTI: diffusion tensor imaging.

Supplementary table 4: MST metrics from studies on neurodegenerative disorders

Modality	***First Author***	**Clinical group**	**Diameter**	**LF**	**Kappa**	**Th**
EEG delta	*Das 2020*	EC MCI-AD	**0.475 ± 0.079**	0.42 ±0.026	N/A	N/S
		EC control	**0.375 ± 0.079**	0.5 ± 0.158	N/A	N/S
		EO MCI-AD	**0.46 ± 0.078**	0.43 ± 0.026	N/A	N/S
		EO control	**0.37 ± 0.095**	0.505 ± 0.174	N/A	N/S
		MAEC MCI-AD	**0.515 ±0.058**	0.440 ± 0.084	N/A	N/S
		MAEC control	**0.375 ±0.079**	0.500 ± 0.158	N/A	N/S
		MAEO MCI-AD	**0.515 ± 0.058**	0.415 ± 0.075	N/A	N/S
		MAEO control	**0.37 ± 0.095**	0.505 ±0.174	N/A	N/S
	Utianski 2016	PD-CN	**0.42**	**0.56**	N/A	N/A
		PD-D	0.41	0.57	N/A	N/A
		PD-MCI	0.42	0.56	N/A	N/A
		Healthy control	**0.44**	**0.54**	N/A	N/A
	Pozar 2020	MCI	**0.2344 ± 0.0326**	**0.5324 ± 0.0314**	**3.0417 ± 0.2523**	0.3895 ± 0.0471
		Healthy control	**0.2010 ± 0.0494**	**0.5720 ± 0.0517**	**3.6837 ± 0.9523**	0.3937 ± 0.0458
EEG theta	*Das 2020*	EC MCI-AD	**0.505 ± 0.093**	0.395 ± 0.072	N/A	N/S
		EC control	**0.375 ±0.079**	0.500 ± 0.158	N/A	N/S
		EO MCI-AD	**0.56 ±0.097**	0.420 ± 0.072	N/A	N/S
		EO control	**0.37 ±0.095**	0.505 ± 0.072	N/A	N/S
		MAEC MCI-AD	**0.505 ±0.072**	0.490 ± 0.184	N/A	N/S
		MAEC control	**0.375 ±0.079**	0.500 ± 0.072	N/A	N/S
		MAEO MCI-AD	**0.525 ±0.058**	0.440 ± 0.077	N/A	N/S
		MAEO control	**0.37 ±0.095**	0.505 ±0.174	N/A	N/S
	Utianski 2016	PD-CN	**0.40**	**0.58**	N/A	N/A
		PD-D	0.40	0.57	N/A	N/A
		PD-MCI	0.41	0.57	N/A	N/A

		Healthy control	**0.41**	**0.568**	N/A	N/A
EEG alpha	*Van Dellen 2015*	DLB	**0.429 ± 0.044**	**0.568 ± 0.049**	N/A	0.392 ± 0.029
		AD	0.412 ± 0.046	0.588 ± 0.057	N/A	0.399 ± 0.034
		Healthy control	0.401 ± 0.052	0.602 ± 0.066	N/A	0.401 ± 0.037
	Yu 2016	AD	**0.3933**	**0.5838**	**2.996**	0.4035
		FTD	0.3823	0.6062	3.222	0.4077
		SCD	0.3769	0.6069	3.159	0.4098
EEG lower alpha	*Das 2020*	EC MCI-AD	**0.477 ±0.077**	0.530 ± 0.223	N/A	N/S
		EC control	**0.375 ±0.079**	0.500 ±0.158	N/A	N/S
		EO MCI-AD	**0.5 2± 0.248**	**0.311 ± 0.041**	N/A	N/S
		EO control	**0.37 ±0.159**	**0.505 ± 0.174**	N/A	N/S
		MAEC MCI-AD	**0.49 ±0.074**	0.450 ±0.078	N/A	N/S
		MAEC control	**0.375 ±0.079**	0.500 ±0.158	N/A	N/S
		MAEO MCI-AD	**0.5 ±0.066**	0.440 ±0.084	N/A	N/S
		MAEO control	**0.37 ±0.095**	0.505 ±0.174	N/A	N/S
	Utianski 2016	PD-CN	0.41	0.59	N/A	N/A
		PD-D	0.43 *a*	0.55 *a*	N/A	N/A
		PD-MCI	0.42	0.56 *a*	N/A	N/A
		control	0.41	0.58	N/A	N/A
EEG upper alpha	*Das 2020*	EC MCI-AD	**0.491 ±0.062**	0.45 ±0.078	N/A	N/S
		EC control	**0.375 ± 0.079**	0.5 ±0.158	N/A	N/S
		EO MCI-AD	**0.55 ± 0.053**	0.44 ±0.174	N/A	N/S
		EO control	**0.37 ± 0.095**	0.505 ±0.084	N/A	N/S
		MAEC MCI-AD	**0.515 ± 0.078**	0.505 ±0.167	N/A	N/S
		MAEC control	**0.375 ± 0.079**	0.500 ±0.158	N/A	N/S
		MAEO MCI-AD	**0.475 ± 0.079**	0.44 ±0.046	N/A	N/S
		MAEO control	**0.37 ± 0.094**	0.505 ±0.174	N/A	N/S
	Utianski 2016	PD-CN	0.40	0.58	N/A	N/A
		PD-D	0.43 *a*	0.57	N/A	N/A

		PD-MCI	0.41	0.57	N/A	N/A
		healthy control	0.41	0.57	N/A	N/A
EEG beta	*Das 2020*	EC MCI-AD	**0.46 ± 0.077**	0.450 ±0.033	N/A	N/S
		EC control	**0.375 ± 0.079**	0.500 ±0.158	N/A	N/S
		EO MCI-AD	**0.46 ± 0.077**	0.430 ±0.026	N/A	N/S
		EO control	**0.37 ± 0.095**	0.505 ±0.174	N/A	N/S
		MAEC MCI-AD	**0.495 ± 0.0831**	0.460 ±0.046	N/A	N/S
		MAEC control	**0.375 ± 0.079**	0.500 ±0.158	N/A	N/S
		MAEO MCI-AD	**0.53 ± 0.042**	0.460 ±0.046	N/A	N/S
		MAEO control	**0.37 ± 0.095**	0.505 ±0.174	N/A	N/S
	Utianski 2016	PD-CN	0.41	0.56	N/A	N/A
		PD-D	0.41	0.57	N/A	N/A
		PD-MCI	0.42	0.56	N/A	N/A
		Healthy control	0.41	0.56	N/A	N/A
EEG gamma	*Das 2020*	EC MCI-AD	0.455 ±0.142	0.365 ± 0.129	N/A	N/S
		EC control	0.425 ±0.130	0.520 ± 0.211	N/A	N/S
		EO MCI-AD	0.455 ±0.142	0.360 ± 0.126	N/A	N/S
		EO control	0.415 ±0.123	0.440 ± 0.084	N/A	N/S
		MAEC MCI-AD	0.465 ±0.111	0.455 ± 0.174	N/A	N/S
		MAEC control	0.41 ±0.158	0.460 ± 0.097	N/A	N/S
		MAEO MCI-AD	0.455 ±0.142	0.360 ± 0.126	N/A	N/S
		MAEO control	0.41 ±0.158	0.490 ± 0.126	N/A	N/S
EEG	*Peraza 2018*	PDD	N/A	N/A	N/A	N/A
		DLB	N/A	N/A	N/A	N/A
		AD	N/A	N/A	N/A	N/A
		Healthy control	N/A	N/A	N/A	N/A
	Fraschini 2016	ALS	N/A*	N/A*	N/A*	N/A*
		Healthy control	N/A*	N/A*	N/A*	N/A*
MEG delta	*Sorrentino 2017*	AGHLS cohort	N/A	0.0515 ± 0.0148 [b]	N/A	0.3840 ± 0.0130

	Olde Dubbelink 2014	PD de novo	N/A	0.604 ± 0.023	N/A	0.437 ± 0.014
		Control	N/A	0.615 ± 0.037	N/A	0.450 ± 0.029
MEG theta	*Sorrentino 2017*	AGHLS cohort	N/A	0.5023 ± 0.0135	N/A	0.3777 ± 0.1309
MEG lower alpha	*Sorrentino 2017*	AGHLS cohort	N/A	0.5506 ± 0.0162	N/A	0.3795 ± 0.0157
MEG upper alpha	*Sorrentino 2017*	AGHLS cohort	N/A	0.5007 ± 0.0147	N/A	0.3758 ± 0.0139
	Olde Dubbelink 2014	PD de novo	N/A	**0.566 ± 0.018**	N/A	**0.423 ± 0.023**
		Healthy control	N/A	**0.594 ± 0.026**	N/A	**0.444 ± 0.023**
MEG beta	*Sorrentino 2017*	AGHLS cohort	N/A	0.5417 ± 0.0174 *b*	N/A	0.3740 ± 0.0154 *b*
MEG gamma	*Sorrentino 2017*	AGHLS cohort	N/A	0.4918 ± 0.0165	N/A	0.3708 ± 0.0138
MEG	*Jacini 2018*	Amnestic MCI	N/A	N/A	N/A*	N/A*
		Healthy control	N/A	N/A	N/A*	N/A*
	Lopez 2017	MCI	N/A*	N/A	N/A*	N/A*
		Healthy control	N/A*	N/A	N/A*	N/A*
	Sorrentino 2018	advanced ALS	N/A	N/A*	N/A	N/A*
		early ALS	N/A	N/A*	N/A	N/A*
		Healthy control	N/A	N/A*	N/A	N/A*
fMRI	*Wang 2018*	AD	N/A*	N/A*	N/A	N/A*
		MCI	N/A*	N/A*	N/A	N/A*
		Healthy control	N/A*	N/A*	N/A	N/A*
fMRI	*Saba 2019*	FTD	N/A	N/A	N/A	N/A
		Healthy control	N/A	N/A	N/A	N/A
DTI	*Jonak 2021*	LHON	0.192 ± 0.051	**0.533 ± 0.081**	N/A	**0.384 ± 0.056**
		Healthy control	0.174 ± 0.33	**0.603 ± 0.039**	N/A	**0.442 ± 0.035**

Data are presented as mean and when available SD. (SE) indicates the use of standard error instead of SD. Bold text represents significant results ($p < 0.05$). For EEG and MEG only frequency bands for which data was available are displayed.N/A: not available; measure not described in study, N/A* measure described in study; data not available, N/S: not significant, EEG: electroencephalography, fMRI: functional magnetic resonance imaging, DTI: diffusion tensor imaging, DLB: dementia with Lewy Bodies, AD: Alzheimer's disease, PD: Parkinson's disease, FTD: Frontotemporal dementia, SCD: subjective cognitive decline, MCI: mild cognitive impairment, EC: eyes closed EEG recording, EO: EO EEG recording, MAEO: mental arithmetic + eyes open EEG recording, MAEC: mental arithmetic + eyes closed EEG recording, ALS: amyotrophic lateral sclerosis, AGHLS: Amsterdam Growth and Health Longitudinal Study, LHON: Leber's hereditary optic neuropathy.

[a] significant compared to PD cognitively normal

[b] significant correlation with Insulin-like Growth Factor (IGF-1) levels

Supplementary table 5: MST metrics from studies on multiple sclerosis

Modality	First Author	Clinical group	Diameter	LF	Kappa	Th
MEG delta	*Tewarie 2015*	MS	0.222 ± 0.009	**0.543 ± 0.016**	N/A	0.411 ± 0.013
		Healthy control	0.218 ± (0.010	**0.556 ± 0.014**	N/A	0.416 ± 0.011
	Tewarie 2014 (a)	MS	N/A*	0.53 ± 0.022	3.1 ± 0.17	0.403 ± 0.023
		Healthy control	N/A*	0.53 ± 0.021	3.16 ± 0.17	0.403 ± 0.021
MEG theta	*Tewarie 2014 (b)*	MS	**0.18 (0.17–0.22)**	**0.62 (0.57–0.68)**	3.69 (3.17–4.36)	N/A (0.39–0.51)
		Healthy control	**0.21 (0.18–0.24)**	**0.59 (0.56–0.65)**	3.6 (3.14–4.44)	0.43 (0.40–0.47)
	Tewarie 2015	MS	**0.229 ± 0.010**	**0.541 ± 0.017**	N/A	0.413 ± 0.015
		Healthy control	**0.214 ± 0.006**	**0.555 ± 0.017**	N/A	0.416 ± 0.017
	Tewarie 2014 (a)	MS	N/A*	0.53 ± 0.024	3.10 ± 0.15	0.403 ± 0.023
		Healthy control	N/A*	0.53 ± 0.024	3.10 ± 0.19	0.405 ± 0.017
MEG lower alpha	*Tewarie 2015*	MS	0.229 ± 0.0105	0.544 ± 0.016 [a]	N/A	0.415 ± 0.013
		Healthy control	0.225 ± 0.010	0.555 ± 0.018 [a]	N/A	0.419 ± 0.013
MEG upper alpha	*Tewarie 2014 (b)*	MS	**0.21 (0.17–0.23)**	**0.59 (0.46–0.62)**	**3.54 (2.54–3.91)**	**0.43 (0.38–0.47)**
		Healthy control	**0.15 (0.12–0.21)**	**0.62 (0.58–0.68)**	**3.7 (3.39–4.92)**	**0.45 (0.42–0.52)**
	Tewarie 2015	MS	0.226 ± 0.010	0.542 ± 0.018	N/A	0.411 ± 0.014 [a]
		Healthy control	0.221 ± 0.013	0.556 ± .019	N/A	0.420 ± 0.011
	Tewarie 2014 (a)	MS	N/A*	**0.530 ± 0.021**	**3.1 ± 0.18**	**0.400 ± 0.019**
		Healthy control	N/A*	**0.540 ± 0.020**	**3.2 ± 0.14**	**0.410 ± 0.021**
MEG beta	*Tewarie 2014 (b)*	MS	0.22 (0.18–0.24)	0.58 (0.53–0.62)	3.33 (2.96–4.65)	0.43 (0.39–0.47)
		Healthy control	0.21 (0.19–0.22)	0.58 (0.56–0.61)	3.41 (3.12–4.32)	0.42 (0.39–0.45)
	Tewarie 2015	MS	0.225 ± 0.010	0.538 ± 0.017	N/A	0.420 ± 0.013
		Healthy control	0.221 ± 0.009	0.548 ± 0.015	N/A	0.414 ± 0.011
MEG gamma	*Tewarie 2015*	MS	0.225 ± 0.009	0.535± 0.018	N/A	0.404 ± 0.013
		Healthy control	0.221 ± 0.008	0.541 ± 0.015	N/A	0.408 ± 0.012
fMRI	*Tewarie 2015*	MS	0.320 ± 0.053	0.320 ± 0.053	N/A	0.270 ± 0.029 [a]
		Healthy control	0.320 ± 0.068	0.330 ± 0.030	N/A	0.260 ± 0.020 [a]

MEG	*Nauta 2020*	MS	N/A*	N/A*	N/A	N/A*
		follow up design	N/A*	N/A*	N/A	N/A*

Data is presented as mean and when available SD, except from Tewarie 2014 (b) where data is presented as median and range (min – max). Bold text represents significant results ($p < 0.05$). For MEG only frequency bands for which data was available are displayed.

N/A: not available; measure not described in study; N/A* measure described in study; data not available, EEG: electroencephalography, fMRI: functional magnetic resonance imaging, DTI: diffusion tensor imaging, MS: multiple sclerosis

[a] reported by Tewarie *et al.* as not significant anymore after correction for multiple comparisons.

Supplementary table 6: MST metrics from studies on epilepsy

6a

Modality	First author	Clinical group	Diameter	LF	Kappa	Th
EEG delta	*Van Diessen* 2016	Chld GE	0.474 ± 0.009 (SE)	0.539 ± 0.009 (SE)	N/A	N/A
		Chld FE	0.490 ± 0.006 (SE)	**0.528 ± 0.006 (SE)**	N/A	N/A
		Healthy Control	0.466 ± 0.003 (SE)	0.547 ± 0.003 (SE)	N/A	N/A
EEG theta	*Van Diessen 2016*	Chld GE	0.471 ± 0.00 (SE)	0.545 ± 0.009 (SE)	N/A	N/A
		Chld FE	0.471 ± 0.006 (SE)	0.547 ± 0.006 (SE)	N/A	N/A
		Healthy Control	0.473 ± 0.003 (SE)	0.548 ± 0.004 (SE)	N/A	N/A
EEG lower alpha	*Van Diessen 2016*	Chld GE	0.464 ± 0.00 (SE)	0.562 ± 0.010 (SE)	N/A	N/A
		Chld FE	**0.456 ± 0.006 (SE)**	**0.569 ± 0.007 (SE)**	N/A	N/A
		Healthy Control	0.479 ± 0.004 (SE)	0.545 ± 0.004 (SE)	N/A	N/A
EEG upper alpha	*Van Diessen 2016*	Chld GE	0.459 ± 0.010 (SE)	0.578 ± 0.011 (SE)	N/A	N/A
		Chld FE	0.454 ± 0.007 (SE)	0.576 ± 0.008 (SE)	N/A	N/A
		Healthy Control	0.458 ± 0.004 (SE)	0.569 ± 0.004 (SE)	N/A	N/A
EEG beta	*Van Diessen 2016*	Chld GE	0.502 ± 0.008 (SE)	0.516 ± 0.009 (SE)	N/A	N/A
		Chld FE	0.494 ± 0.005 (SE)	0.516 ± 0.006 (SE)	N/A	N/A
		Healthy Control	0.494 ± 0.003 (SE)	0.523 ± 0.003 (SE)	N/A	N/A
EEG	*Van Diesen 2014*	Chld. suspected new onset Focal epilepsy	N/A*	N/A*	N/A	N/A
		Healthy control (sleep deprivation design)	N/A*	N/A*	N/A	N/A

6b

Modality	First author	Clinical group	Diameter	LF	Kappa	Th
EEG theta	*Fraschini 2014*	Nonresponder pre-VNS	**0.404 ± 0.023**	0.606 ± 0.034	N/A	0.407 ± 0.030
		Nonresponder post-VNS	**0.463± 0.029**	0.544 ± 0.064	N/A	0.380 ± 0.053
		Responder pre-VNS	**0.433 ± 0.046**	0.581 ± 0.081	N/A	0.396 ± 0.064
		Responder post-VNS	**0.359 ± 0.045**	0.667 ± 0.067	N/A	0.442 ± 0.036
MEG alpha	*Van Dellen 2014*	Meningioma, with seizures				
		Post operative seizures (baseline)	0.2069 ± 0.0148	0.5907 ± 0.0180	3,5377 ± 0,1818	N/A
		Post operative seizure free (baseline)	0.2198 ± 0.0241	0.5673 ± 0.033	3,3435 ± 0,3044	N/A
		Post operative seizures (9-15 month follow up)	0.2139 ± 0.0170	0.5859 ± 0.0226 $	3,6055 ± 0,3647	N/A
		Post operative seizure free (9-15 month follow up)	0.2092 ± 0.0228	0,5932 ± 0,0383 $	3,5417 ± 0,3882	N/A
fMRI	*DeSalvo 2020*	TLE pre-op, seiz post-op	N/A	N/A*	N/A	N/A*
		TLE pre-op, seiz free post-op (n=22)	N/A	N/A*	N/A	N/A*

6c

Modality	First author	Clinical group	Diameter	LF	Kappa	Th
EEG	*Kinney-Lang 2019*	Chld early onset epilepsy				
		Cognitively normal	N/A*	N/A*	N/A	N/A
		Mild cognitive impairment (n=7)	N/A*	N/A*	N/A	N/A
		Severe cognitive impairment (n=13)	N/A*	N/A*	N/A	N/A

Segment *a* represents studies assessing disease effect. Segment *b* represents studies assessing treatment effect. Segment *c* contains one study assessing the correlation between MST metrics and cognitive scores.

Data are presented as mean and when available SD. Bold text represents significant results ($p < 0.05$). For EEG and MEG only frequency bands for which data was available are displayed. N/A: not available; measure not described in study, N/A* measure described in study; data not available, EEG: electroencephalography, fMRI: functional magnetic resonance imaging, Chld GE: Drug naive children with generalized epilepsy, Chld FE: Drug naive children with focal epilepsy, TLE: temporal lobe epilepsy, VNS: Vagal nerve stimulation.

[$]: significant difference compared to seizure free group at 9-15 months after surgery.

Supplementary table 7: MST metrics from studies on other disorders

Modality	First Author	Clinical group	Diameter	LF	Kappa	Th
EEG delta	*Numan 2017*	Hypo-active delirium	0.481 (0.449-0.493)	0.535 (0.504-0.579)	N/A	0.376 (0.355-0.408)
		Recovery anesthesia	**0.457 (0.438-0.469)**	**0.558 (0.532-0.594)**	N/A	0.385 (0.366-0.402)
		Non-delirious control	0.486 (0.446-0.516)	0.522 (0.500-0.563)	N/A	0.371 (0.356-0.388)
EEG alpha	*Numan 2017*	Hypo-active delirium	0.497 (0.454-0.516	**0.513 (0.485-0.547)**	N/A	0.370 (0.350-0.398)
		Recovery anesthesia	0.467 (0.442-0.496)	0.534 (0.493-0.563)	N/A	0.368 (0.346-0.394)
		Non-delirious control	0.469 (0.438-0.500)	0.545 (0.516-0.574)	N/A	0.377 (0.363-0.393
EEG beta	*Numan 2017*	Hypo-active delirium	0.522 (0.485-0.563)	0.485 (0.450-0.531)	N/A	0.351 (0.330-0.370)
		Recovery anesthesia	0.512 (0.485-0.547)	0.511 (0.488-0.532)	N/A	0.372 (0.360-0.389)
		Non-delirious control	0.510 (0.485-0.543)	0.501 (0.469-0.532)	N/A	0.360 (0.343-0.382)
MEG	*Nieboer 2020*	Migraine	N/A*	N/A*	N/A*	N/A*
		subgroup recently diagnosed	N/A*	N/A*	N/A*	N/A*
		subgroup long-term	N/A*	N/A*	N/A*	N/A*
		Healthy controls	N/A*	N/A*	N/A*	N/A*
MEG	*Van Nieuwenhuizen 2018*	meningioma patients	N/A	N/A*	N/A	N/A*
		Healthy controls	N/A	N/A*	N/A	N/A*
fMRI	*Van Montfort 2018*	Delirium	**0.30 ± 0.05**	**0.32 ± 0.04**	2.38 ± 0.11	0.26 ± 0.02
		Post Delirium	0.30 ± 0.06	0.36 ± 0.04	2.45 ± 0.11	0.31 ± 0.03
		Non delirious control	0.28 ± 0.04	0.35 ± 0.03	2.41± 0.07	0.28 ± 0.02

Data is presented as mean and when available SD. For Numan *et al* data is presented as median and Interquartile range. Bold text represents significant results ($p < 0.05$). For EEG and MEG only frequency bands for which data was available are displayed. For EEG and MEG only frequency bands for which data was available are displayed. N/A: not available; measure not described in study, N/A* measure described in study; data not available, NS: not significant, EEG: electroencephalography, fMRI: functional magnetic resonance imaging.